

TABLE OF CONTENTS
S 630 - ROAD, PARKING LOT AND MISCELLANEOUS PAVEMENT
WORKS

	Page No.
1. SCOPE OF WORK	S 630-2
2. SUBGRADE PREPARATION FOR ROADS AND PARKING LOTS	S 630-2
3. ROAD WORK	S 630-3
3.1 General	S 630-3
3.2 Materials	S 630-3
3.3 Contruction Requirements.....	S 630-5
3.4 Testing of Hot Asphalt Concrete Paving	S 630-7
4. PARKING LOT	S 630-7
4.1 Materials	S 630-7
4.2 Construction Requirements.....	S 630-8
5. CONCRETE WALKS, DRIVES AND INCIDENTAL PAVING	S 630-10
5.1 General.....	S 630-10
5.2 Materials	S 630-10
5.3 Construction Requirements.....	S 630-10
6. CURBS AND GUTTERS.....	S 630-11
6.1 General.....	S 630-11
6.2 Materials	S 630-12
6.3 Construction Requirements.....	S 630-12
7. WHEEL STOPS.....	S 630-13
7.1 General.....	S 630-13
7.2 Materials	S 630-13
7.3 Contruction Requirements	S 630-13
8. ROAD MARKING	S 630-13
9. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)	S 630-14

S 630 - ROAD, PARKING LOT AND MISCELLANEOUS PAVEMENT WORKS

1. SCOPE OF WORK

The scope of work under this section shall cover the furnishing of all materials, labor, equipment's, transportation, and installation of the following works and as indicated in the Technical Specification S-001.

- 1.1 Preparing the subgrade shall be followed by placing the crushed stone subbase and base courses asphalt primer for all roads, temporary and permanent, and in the case of permanent roads placing the surface course including the asphalt concrete wearing course.
- 1.2 Preparing the subgrade is based course (CTB) and sand bedding, and laying concrete blocks for parking lots.
- 1.3 Constructing concrete walks, drives and incidental concrete paving slabs.
- 1.4 Curbs and gutters.
- 1.5 Wheel stops at parking areas.
- 1.6 Road marking

2. SUBGRADE PREPARATION FOR ROADS AND PARKING LOTS

- 2.1 Excavate subgrade or fill, as required, and then grade and compact to provide a firm foundation of uniform density throughout. Completed subgrade shall be true to alignment, grade and cross-section, including the required crown.
- 2.2 CBR test on the finished level of complete subgrade shall have a minimum value of 6% in the 98% dry density compacted condition, as per ASTM D698, Method D.
- 2.3 Prior to placing of any fill required to bring the subgrade to its proper level, any loose or otherwise unapproved fill materials, organic materials, or any foreign or deleterious matter shall be removed.
- 2.4 Fill holes, ruts and similar defects with approved fill materials. Unstable areas, projecting stone or rock, and similar defects shall be cut out and the areas shall be filled. Round shaped gravel shall categorically not be used.

- 2.5 All fill required shall be completed as specified in Specification S-300 “Earth Works”.
- 2.6 Following the completion of subgrade preparation, the Bidder shall proceed with the next phase of work as specified in Clause 3.
- 2.7 Drainage culverts under the subgrade shall have been constructed before the subgrade work is carried out.
- 2.8 The Bidder shall provide temporary drainage as required to facilitate his works activities during the wet season.

3. ROAD WORK

3.1 General

The Bidder shall perform all phase of road construction work per the requirements stated in this Specification and in accordance with the approved Bidder’s drawings. Roads shall be categorized as below:

- Type T1: Temporary road without surface course, to be removed prior to the completion of the Project.
- Type T2: Temporary road without surface course during the construction phase of the Project, to convert to permanent road by receiving the Base Course and surface course at a later stage.(Rehabilitation or improved Base Course)
- Type P: Permanent road without any apparent interruption between the placement of the base and surface courses.

3.2 Materials

Materials used for the subbase, base and surface courses shall consist of materials as specified below:

3.2.1 Subbase Course Material for Types T1, T2 and P Roads

Materials shall be obtained from a source approved by the Owner. This material shall consist of well graded crushed stone with hard, strong, durable rock pieces having the following gradation (size C-40, JIS A 5001), or approved equivalent.

Name of Crushed Stone; Crushed Run,C-40

Sieve Size (mm)	Percentage Passing by Weight (%)
53	100
37.5	95~100
19	50~80
4.75	15~40
2.36	5~25

3.2.2 Base Course Material for Types T1, T2 and P Roads

This material shall consist of well graded crushed stone with hard, strong, durable pieces having the following gradation (size M-40, JIS A 5001), or approved equivalent.

Name of Crushed Stone; Mechanical Stabilization,M-40

Sieve Size (mm)	Percentage Passing by Weight (%)
53	100
37.5	95~100
19	60~90
4.75	30~65
2.36	20~50
0.425	10~30
0.075	2~10

3.2.3 Surface Course Material (Hot Asphalt Concrete Paving) for Types T2 and P Roads

Materials for asphalts paving shall conform to the following, or approved equivalent.

Asphalt primer	ASTM D2027, Grade MC-70
Asphaltic concrete	ASTM D3515, Grade MC-70, job mix formula acceptable to the Owner, design density between 94 and 97 percent density of void-less mixture determined by Hubbard-field test or ASTM D1074.
Asphalt cement	ASTM D946, Penetration Grade 85-100 or AASHTO M20, Penetration Grade 85-100.
Coarse aggregate	ASTM D692, Size No. 67 or 68.
Fine aggregate	ASTM D1073, Grading No. 1; add mineral filler as required to provide at least 5

percent of total aggregate passing No. 200 sieve for surface courses.

Mineral filler

Lime stone dust or portland cement.

3.3 Construction Requirements

3.3.1 General

The works specified in Clause 3.3.2, 3.3.3, and 3.3.4 shall be carried out in sequence without any undue interruption between the individual courses, for both temporary and permanent roads. The requirements of Clause 3.3.5 shall be followed for the completion of permanent road work.

3.3.2 Construction of Subbase Course (30 cm thick)

- 1) Immediately after placing the subbase material, the surface shall be proof rolled and compacted to a dry density not less than 98 percent of the maximum dry density as determined by the method specified in ASTM D698, Method D.
- 2) Any soft and unsound areas, encountered during the preparation and compaction of the surfaces, shall be removed. The unsound areas, such as pockets or cavities shall be excavated and backfilled with approved material to the same depth, and beyond the neat lines and elevations as shown on the approved Bidder's drawings or as directed by the Owner.
- 3) Shaping and compaction shall be carried out concurrently so that upon completion, the surfaces shall be smooth and everywhere within plus or minus 30 mm of the required grade, but not uniformly high or low.
- 4) Any surface if disturbed for any reason before or during the placing of the base course material shall be reworked and re-compacted at the Bidder's expense.
- 5) Minimum CBR for Sub Base Course to be 30%

3.3.3 Construction of Base Course (20 cm thick)

- 1) After the subbase course for the road has been completed, the Bidder shall place the base course, in layers not less than 100 mm of compacted thickness. (Two layers: 2 x 100 mm = 200 mm)

- 2) Road bases and surfaces shall be compacted to a dry density not less than 98 percent of the maximum dry density as determined by the method specified in ASTM D698, Method D. Water shall be added during compaction as required to achieve the specified dry density.
- 3) Road surfaces shall be shaped and compacted to the lines, grades and dimension shown on the approved Bidder's drawings by equipment designed for that purpose. Such shaping and compaction shall be carried out concurrently so that upon completion the surface shall be smooth and everywhere within plus or minus 15 mm of the required grade, and shall be compacted to the degree specified for the full depth. Any local depressions which do not meet the above tolerances shall be filled with approved granular surface materials.
- 4) Minimum CBR for Base Course to be 80%

3.3.4 Asphalt Primer/Prime Coat

- 1) The crushed rock base course shall be dried, swept clean of all dust, dirt and loose particles.
- 2) A prime coat of asphalt prime ASTM D2027 GRADE MC70 shall be sprayed uniformly over the base course surface at a rate of 0.90 liter per square meter as required to fill all voids and interstices.
- 3) The asphalt prime coat shall be placed when the humidity, air temperature and general conditions, and the condition of the compacted base course layer are acceptable, as directed by the Owner.

3.3.5 Hot Asphalt Concrete paving (7.5 cm thick)

- 1) Asphaltic concrete shall be placed when the humidity, air temperature, general weather conditions, and the condition of the compacted base course are suitable, as directed by the Owner. This shall apply to both Type T2 and P roads.
- 2) The paving shall be laid in two layers of 37.5 mm thick (compacted) and shall be laid hot using a mechanical spreader having a width between 2.4 and 3.7 m. The work shall be laid out so that lanes of equal width are used throughout area being surfaced. The spreader shall be equipped with a screed or strike-off device which produces a smooth, uniform surface of proper shape and grade.
- 3) The asphalt paving shall be compacted continuously as closely behind the paver as necessary to obtain the specified density without causing undue displacement. The completed paving shall have a density minimum 95 per cent of the density of a laboratory specimen made in the proportions of the job mix formula.

- 4) Care shall be taken in compact the surface course along the entire length of the pavement edges. Before it is compacted, the material along the unsupported edges shall be slightly elevated with a tamping tool. This will permit the full weight of the roller wheel to bear on the material at the extreme edges of the pavement. In rolling pavement edges, roller wheels shall extend 50 to 100 mm beyond the pavement edge. Adequate precaution shall be taken to prevent distortion of the pavement edge from the specified line and grade.
- 5) Curbs, sidewalks, buildings, and other surfaces shall be protected from damage by paving equipment and from deposits of asphalt materials. Deposits of asphalt materials on such surfaces shall be considered permanent damage and shall be repaired by the Bidder as directed by the Owner at no cost to the Owner.

3.4 Testing Of Hot Asphalt Concrete Paving

- 3.4.1 The job mix formula, aggregate requirements, and compaction density requirements for asphalt concrete paving shall be determined prior to the start of the paving work by an independent testing laboratory. The laboratory shall be retained by the Bidder throughout the Project.
- 3.4.2 If the original density tests indicate the compaction to be inadequate, additional compaction and testing shall be performed until the density meets the requirements of this Specification and acceptable to the Owner.

4. PARKING LOT

4.1 Materials

Materials shall conform to the following:

- 4.1.1 The Base Course thickness to be 20 cm.

4.1.2 Concrete Block Paving (Interlocking)

The nominal size of each block shall be 210 mm long x 105 mm wide 100 mm high.

The average compressive strength of block shall be not less than 49 N/mm²; however, no individual block strength shall be below 40 N/mm².

4.1.3 Sand Bedding

Laying coarse material shall be natural or crushed rock fines and passing the 5mm BS sieve and shall have the following grading:

Sieve Size	Percentage Passing by Weight (%)
5mm	90~100
2.36mm	75~100
1.18mm	55~90
600 μ m	35~90
300 μ m	8~30
150 μ m	0~10

4.1.4 Sand Filter

The natural sand filter shall be grades with maximum particle size no greater than 1.18 mm and containing approximately 10 % by weight passing the 75 microns sieve.

4.2 Construction Requirements

4.2.1 Sand Bedding Layer

1) Laying Course

The laying course shall be such that, after compaction, it forms a nominal uniform 50 mm thick layer under the blocks.

- 2) This layer shall be lightly compacted by means of the vibrating plate compactor similar to that used for compacting the block paving course. A further layer of loose material shall be spread and screened to create a loose surface on to which the block can be placed
- 3) Due allowance shall be made in setting up the screeding rails for the subsequent compaction of the laying course. A small trial area shall be carried out to the determined the allowance.
- 4) Care shall be taken to avoid localized disturbance of the prepared laying course sand by pedestrian or wheeled traffic prior to placing blocks. The area of laying course prepared shall be such that the position of its boundary is more than one meter from the position of the laying face at the end of the working period wherever practicable.

4.2.2 Laying Concrete Block Paving

- 1) At the perimeter of all areas to be paved with concrete block will be an edge restraint of curbing, channel block, drainage channel wall or similar. These shall be formed prior to laying concrete blocks to prevent the loosening of the laying course material and shall be sufficiently cured as not to be adversely affected by the compactable vibration of the surface course.

- 2) Laying of blocks shall not be permitted during periods of rain that will affect the laying course, or jeopardize the integrity of any part of the pavement.
- 3) The blocks shall be laid in hand tightly, in a pattern, working from an existing laying face edge or edge restraint wherever possible. Mechanical force shall not be used to obtain tight joints. Whole blocks shall be laid first, then closure units shall be laid. The area to be laid shall be complete as far as possible in whole block units. Wherever possible, infilling to boundaries and obstructions shall proceed as laying of the surface course proceeds and, in any case, infilling shall be completed before compaction commences.
- 4) Where trimming of blocks is necessary, portions of less than a quarter of a whole block shall not be permitted.

4.2.3 Compaction

- 1) The surface course shall be compacted by approval plate compactors in order to ensure the filling of the lower portion of the block-to-block joint by the laying materials. Two or three passes of the compaction equipment will normally be required to achieve this condition.
- 2) The plate compactor shall have plan area of not less than 0.25 m². The compactor shall transmit an effective force of 75-100 kN/m² of plate area, the frequency of vibration shall be within the range of 75-100 Hz.
- 3) Compaction shall follow laying as soon as possible but shall not be carried out within one meter of the laying face. Apart from this edge strip no area of paving shall be left un-compacted at the completion of the day's work.
- 4) After compaction of the surface course, dry sand or crushed rock fines shall be spread over the surface and brushed into the joints. The materials shall be graded with maximum particle size no greater than 1.18 mm and containing approximately 10 % by weight passing the 75 micron sieve.
- 5) Care shall be taken to avoid the use of a material which might stain the pavement surface, and in addition the material shall not contain any deleterious salts or other contaminations.
- 6) The block paved area shall then be re-vibrated as before in order to encourage the filling of the upper part of the block-to-block joint by the surface-applied material. Top filling and final compaction shall be completed as soon as practicable after laying and in any case prior to the termination of work on that day.

5. CONCRETE WALKS, DRIVES AND INCIDENTAL PAVING

5.1 General

This work shall consist of the construction of concrete walks, drives and incidental paving slabs, in accordance with this Specification and in reasonable close conformity with the lines and grades as indicated on the approved Bidder's drawing or as directed by the Owner.

5.2 Materials

Materials shall conform to Specification S-340 "Concrete Works" unless otherwise specified. All concrete shall be GRADE 25.

5.3 Construction Requirements

5.3.1 Excavation

Excavation shall be made to the required depth and to a width that will permit the installation and bracing of the forms. The subgrade shall be shaped and compacted to a firm, even surface conforming to the section shown on the approved Bidder's drawings. All soft and yielding material shall be removed and replaced with the approved material at the Bidder's expense. Excavation and fill work shall conform to the requirements of Specification S-300 "Earth Work".

5.3.2 Forms

Forms shall be wood or metal and shall extend for the full area of the concrete. All formworks shall conform to the requirements of Specification S-340 "Concrete Works".

5.3.3 Subgrade

The subgrade shall be thoroughly moistened immediately prior to the placing of the concrete.

5.3.4 Depositing and Finishing

The concrete shall be deposited on the moist subgrade and struck off to the required thickness. It shall be temped sufficiently to bring the mortar to the surface. The surface shall be finished with a wood float or steel trowel, provided that the surface shall be finally brushed in order to leave a slightly rough finish. All joints and edge shall be rounded with an edging tool having a 6 mm radius.

5.3.5 Joints

- 1) Expansion joints shall be of the dimensions specified, and shall be filled with pre-molded expansion joint filler. Dummy joints shall be formed by a jointing tool or other acceptable means as directed. The dummy joints shall extend into the concrete for the least 1/4 of the depth and shall be approximately 3 mm wide.
- 2) Construction joints shall be formed around all appurtenances such as manholes, utility poles, etc, extending into and through the paving. Unless otherwise specified, 6 mm pre-molded expansion joint filler shall be installed in these joints. Expansion joint filler indicated shall be installed between concrete paving and any fixed structure such as a building or bridge. This expansion joint material shall extend for the full depth of the paving.
- 3) Unless otherwise specified, expansion joint filler shall be 12 mm thick and shall be installed every 30 m or fraction thereof, between intersecting paving and any fixed structure such as a building, bridge or cubing. This expansion joint material shall extend for the full depth of the paving.
 1. Walks: Unless otherwise directed, the spacing of dummy joints for walks shall be equal to the width of the walk.
 2. Drives: A longitudinal dummy joint shall be formed along the centerline of drives more than 4.5 m wide, and transverse dummy joints shall be formed in drives at not more than 4.5 m intervals.
 3. Incidental Paving: Unless otherwise directed, dummy joints for incidental paving shall be formed at intervals not exceeding 6.0 m in length or width, and incidental paving poured adjacent to existing jointed concrete shall be jointed to match the joints in the existing concrete, with intermediate joints formed as necessary not to exceed the 6.0 m maximum joint spacing.

5.3.6 Curing

Concrete shall be cured for at least 72 hours. Curing shall conform to the requirement of Specification S-340 "Concrete Works".

6. CURBS AND GUTTERS

6.1 General

This work shall consist of the construction of curbs and gutters as shown on the Bidder's approved drawings in accordance with this Specification and in reasonably close conformity with the lines, grades, dimensions and typical

sections as indicated on the approved Bidder's drawings or as directed by the Owner.

6.2 Materials

Concrete for concrete curbs and gutters shall be GRADE 25 conforming to the requirements of Specification S-340 "Concrete Work".

6.3 Construction Requirements

6.3.1 Subgrade

The subgrade shall be shaped to the required depth below the finished surface in accordance with the dimensions shown on the approved Bidder's drawing and shall be compacted to a firm, even surface. When possible, the subgrade shall be shaped and compacted at the same time and in the same manner as the subgrade for the pavement slab. All soft and yielding spots or any unstable material encountered shall be removed and replaced with suitable material, at the Bidder's expense, and be thoroughly compacted. When foundation and underground pipes or culverts are to be placed under curbing and gutters, the excavation and backfilling for same shall be completed and compacted before the subgrade for curbing and gutters is prepared.

6.3.2 Concrete Forms

The forms for the curbing or gutter shall be of wood or metal, straight, free from warp and of sufficient strength when staked to resist the pressure of the concrete without springing. All forms shall be cleaned thoroughly and greased or soaped before concrete is placed against them. Forms which have become worn, bent or broken shall not be used until satisfactorily repaired and straightened. Repaired forms shall not be used until inspected and approved by the Owner. An approved mechanical curb forming machine may be used without forms.

6.3.3 Concrete Joints

- 1) Joints shall be formed in continuous curbing to correspond with transverse joints in the pavement slab. All joints shall extend entirely under the curb section and through the curb and shall be finished and filled with prescribed filler.
- 2) All other types of curbing shall be provided with 6 mm joint at intervals of 6 m, unless otherwise indicated on the approved Bidder's drawing, except where shorter sections are necessary for closures. The separation shall be effected by using steel plates 6 mm in thickness, cut to the true section, and set vertically in the forms until the concrete has set sufficiently to permit withdrawal of the plates.

6.3.4 Backfilling

After the curb or gutter has been set sufficiently, the Bidder shall backfill adjacent to the curb or gutter with the approved material. This material shall be thoroughly compacted in layers of not over 150 mm thicknesses.

7. WHEEL STOPS

7.1 General

This work shall consist of the construction of wheel stops as shown on the approved Bidder's drawings in accordance with this Specification and in reasonably close conformity with the dimensions and typical sections as indicated on the approved Bidder's drawings or as directed by the Owner.

7.2 Materials

Concrete for wheel stops shall be precast concrete, GRADE 25 conforming to the requirements of Specification S-340 "Concrete Works".

7.3 Construction Requirements

The forms for the wheel stop shall be of wood or metal, straight, free from warp and of sufficient strength when staked to resist the pressure of the concrete without springing. All forms shall be cleaned thoroughly and greased or soaped before concrete is placed against them. Forms which have become worn, bent or broken shall not be used until satisfactory repaired and straightened. Repaired forms shall not be used until inspected and approved by the Owner. An approved mechanical wheel stop forming machine may be used without forms.

8. ROAD MARKING

8.1 The Bidder shall design and provide road marking, unless otherwise directed by the Owner.

8.2 Road marking shall conform to the standards of Class 3, No.1, JIS K 5665 [Traffic Paint (Melt Application)] or approved equivalent.

8.3 During execution of road marking, moisture, dirt and dust shall be removed from the surface of the pavement. Road marking shall be done carefully in a manner that materials will adhere uniformly to the surface.

9. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

- 9.1 Quality control shall conform to the requirements of Part 4, Subsection 2.7.
- 9.2 The quality control procedures and documentation.

(Last page of S-630)